



Ezaldin Nasan (Attempt 1 of 1)

Test Information

QUESTION 1: CALCULATED NUMERIC

0

out of 20 points

Cold liquid **LPG** (Liquefied Petroleum Gas) at pressure $P_1=10$ bar and temperature $T_1=5^\circ\text{C}$, with molar composition: Propane=0.60, n-Butane=0.3 and i-Butane=0.1, is throttled adiabatically down to a pressure of $P_2=1$ bar. Calculate the Lost Work of the throttling operation, if the flow rate of **LPG** is 100 kmol/hr. Assume that the thermal properties of the **LPG** are described by the **Peng-Robinson** equation of state with Binary Interaction Parameters (BIPs) all set equal to zero. Express your answer in **kW** and round your numerical result into **a single decimal digit**.



Given Answer: ✖ 11.3

Correct Answer: ✔ 12.6

Answer range +/- 1 (11.6 - 13.6)

QUESTION 2: CALCULATED NUMERIC

10

out of 10 points

Calculate the fugacity of dry ice (solid CO_2) at the triple point ($T=216.55\text{ K}$, $P=5.12673\text{ bar}$). Assume that the thermal properties of the vapor phase of carbon dioxide(CO_2) are described by the **Peng-Robinson** equation of state. Express your result in **bar** and round your numerical answer into **a single decimal digit**.



Given Answer: ✔ 4.8

Correct Answer: ✔ 4.8

Answer range +/- 0.1 (4.7 - 4.9)

QUESTION 3: CALCULATED NUMERIC

20

out of 20 points

A liquid solution of n-Pentane (C_5H_{12}) [Component **1**], n-Hexane (C_6H_{14}) [Component **2**] and Cyclohexane (C_6H_{12}) [Component **3**] is initially at $T=25^\circ\text{C}$ and $P=2$ bar. The molar composition of the mixture is $[Z_1, Z_2, Z_3]=[0.40, 0.15, 0.45]$ and the volumetric flow rate 1000 gpm. The mixture is subsequently heated(isobarically) until it reaches its boiling point in a heat exchanger using saturated steam (with quality equal to one) at $P=2.0$ bar. Calculate the minimum steam flow rate required for the heating process. Assume that both phases (vapor and liquid) of solution follow the **Peng-Robinson** Equation of State with Binary Interaction Parameters (BIPs) all set equal to zero. Express your result in **MMSCFD** and round your numerical answer into **two decimal digits**.



Given Answer: ✔ 8.35

Correct Answer: ✔ 9.15

QUESTION 4: CALCULATED NUMERIC

0

out of 10 points

Calculate the **Internal Energy** of the Saturated Vapor of n-Octane in **kJ/mol** at $T=180^{\circ}\text{C}$. The reference state is $h = 0$ at the normal boiling point for saturated liquid. Round your numerical answer **into a single decimal digit**. Assume that both the vapor and liquid phases of n-Octane follow the **Peng-Robinson** Equation of State.



Given Answer: ✖ 41.5

Correct Answer: ✔ 44.1

Answer range +/- 0.1 (44.0 - 44.2)

QUESTION 5: CALCULATED NUMERIC

20

out of 20 points

A fluid follows the following equation of state: $\left(P - \frac{a}{V^2}\right)V = RT$ where a is a constant that depends on the critical properties of the fluid. Calculate the Residual Gibbs Free Energy of this fluid at $T=30^{\circ}\text{C}$ and $P=10$ bar, if the compressibility factor at these conditions is $Z=0.88$.



Express your result in **kJ/kmol** and round your numerical answer **into whole number**.

Given Answer: ✔ -283

Correct Answer: ✔ -283

Answer range +/- 1 (-284 - -282)

QUESTION 6: CALCULATED NUMERIC

20

out of 20 points

Dry Air (with molar composition 21% O_2 and 79% N_2) is compressed from $P_1=1$ bar and $T_1=30^{\circ}\text{C}$ to a final pressure $P_{\text{final}}=100$ bar in a two stage compressor with intermediate cooling of the first stage discharge stream down to $T_3=30^{\circ}\text{C}$ before entering the second stage. The two stages have the same pressure ratio and isentropic efficiency. A thermocouple measures the temperature of the second stage discharge stream to be equal to $T_{\text{final}}=365^{\circ}\text{C}$. The molar flow rate of the Dry Air is 1.5 kmol/s and its thermal properties are described by the **Peng-Robinson** Equation of State with all Binary Interaction Parameters (BIPs) set equal to zero. Calculate the total power requirement for this process. Express your result in **MW** and round your numerical answer **into a single decimal digit**.



Given Answer: ✔ 30

Correct Answer: ✔ 30

Answer range +/- 1 (29 - 31)

FEEDBACK AND NOTES FOR ATTEMPT